

Jarad B. Niemi

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Education

Ph.D. Statistical Science, Duke University, 2009
Thesis: Bayesian Analysis and Computational Methods for Dynamic Modeling
Advisor: Mike West

M.S. Biostatistics, University of Minnesota, 2005
Thesis: Identifying and Evaluating Contrarian Strategies for NCAA Tournament Pools
Advisor: Brad Carlin

B.ChE. Chemical Engineering, University of Minnesota, 1999

Academic Positions

Data Science Program Director	Iowa State University 2024–present
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Statistics Professor Associate Professor Assistant Professor	Iowa State University 2023–present 2017–2023 2011–2017
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Statistics and Actuarial Sciences Visiting Scholar	Simon Fraser University 2019–2020
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Statistics & Applied Probability Assistant Professor	University of California, Santa Barbara 2009–2011
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Employment

Consultant	Eocene Environmental Group, Inc.	2024–2026
	Kemin Industries	2024
	Boehringer Ingelheim Vetmedica, Inc.	2012–2013
	Natural Resources Research Institute	2009–2010
	Denver Health	2008
	Purdue Pharma LP	2007–2008
Research Assistant	Duke University (Mike West)	2006–2009
	University of Minnesota (Brad Carlin)	2004–2005
	University of Minnesota (Grace Peng)	2003–2004
Junior Scientist	Natural Resources Research Institute	2003
	University of Minnesota	2001–2004
Research Engineer	Procter & Gamble	1999–2001

Publications

* Indicates a student advisee or co-advisee.

Refereed Journal Articles

Spencer Wadsworth* and **Jarad Niemi**. (2026) Quantile Forecast Matching with a Bayesian Quantile Gaussian Process Model *Statistics and Computing*, **36**:116. [url](#)

Spencer Wadsworth* and **Jarad Niemi**. (2026) Forecasting Influenza Hospitalizations Using a Bayesian Hierarchical Nonlinear Model with Discrepancy *to appear in Bayesian Analysis* [url](#)

Alex Cleveringa, **Jarad Niemi**, and Fernando Miguez. (2025) Losing Control: What happens when the control treatment is excluded in nitrogen fertilizer rate trials? *Agronomy Journal*, **117**(6):e70243. [url](#)

Jessica A. Nelson, Matt Liebman, **Jarad Niemi**, Richard Cruse, John Tyndall, Chris Witte, Dave James, and Matt Helmers. (2024) The Influence Of Prairie Strips Sown In Midwestern Corn And Soybean Fields On Sediment Discharge Throughout The Year. *Journal of Soil and Water Conservation*, **79**(2), 87–98. [url](#)

Sarabeth Mathis, Alexander E. Webber, Tomas M. Len, Erin L. Murray, Monica Sun, Lauren A. White, Logan C. Brooks, Alden Green, Addison J. Hu, Daniel J. McDonald, Roni Rosenfeld, Dmitry Shemetov, Ryan J. Tibshirani, Sasikiran Kandula, Sen Pei, Jeffrey Shaman, Rami Yaari, Teresa K. Yamana, Pulak Agarwal, Srikar Balusu, Gautham Gururajan, Harshavardhan Kamarthi, B. Aditya Prakash, Rishi Raman, Alexander Rodriguez, Zhiyuan Zhao, Akilan Meiyappan, Shalina Omar, Prasith Baccam, Heidi L. Gurung, Steve A. Stage, Brad T. Suchoski, Marco Ajelli, Allisandra G. Kummer, Maria Litvinova, Paulo C. Ventura, Spencer Wadsworth*, **Jarad Niemi**, (2024) Evaluation of FluSight Influenza in 2021-22 and 2022-23 Seasons with a New Forecasting Target: Laboratory-Confirmed Influenza Hospitalizations. *Nature Communications*, **15**(1), 6289. [url](#)

Velma Lopez, Estee Y Cramer, Robert Pagano, John M Drake, Eamon B O’Dea, Benjamin P Linas, Turgay Ayer, Jade Xiao, Madeline Adee, Jagpreet Chhatwal, Mary A Ladd, Peter P Mueller, Ozden O Dalgic, Johannes Bracher, Tilmann Gneiting, Anja Muhlemann, **Jarad Niemi**, (2024) Challenges of COVID-19 Case Forecasting in the US, 2020-2021. *PLOS Computational Biology*, **20**(5), e1011200. [url](#)

Qinglong Tian, Colin Lewis-Beck*, **Jarad Niemi**, and William Meeker. (2024) Specifying prior distributions in reliability applications. *Applied Stochastic Models in Business and Industry*, **40**(1), 5–62. [url](#)

Cole Dutter, Luis A. Damiano*, **Jarad Niemi**, Bradley A. Miller, Lisa A. Schulte, Matt Liebman, Matt Helmers, Rick Cruse, and Marshall D. McDaniel. (2023) Contour prairie strips affect adjacent soil but have only slight effects on crops. *Field Crops Research*, **296**. [url](#)

Zachary P. Simpson, Jim Jordahl, Andrea Leptin, Fernando E. Miguez, **Jarad Niemi**, Lisa A. Schulte, Michael L. Thompson, Sebastian H. Villarino, and Marshall D. McDaniel. (2023) No - tillage does not on average reduce soil carbon storage compared to conventional tillage. Comment on “Declines in soil carbon storage under no tillage can be alleviated in the long run” by Cai et al. *Geoderma*, **430**. [url](#)

Matt Nowatzke, Luis Damiano*, Fernando Miguez, Gabe McNunn, **Jarad Niemi**, Lisa Schulte-Moore, Emily Heaton, Andy VanLoocke. (2022) Augmenting agroecosystem models with remote sensing data and machine learning changes overall estimates of nitrate-nitrogen leaching. *Environmental Research Letters*, **17**:11, 114010. [url](#)

Estee Cramer, Yuxin Huang, Yijin Wang, Evan Ray, Matthew Cornell, Johannes Bracher, Andrea Brennen, Alvaro Castro Rivadeneira, Aaron Gerding, Katie H House, Dasuni Jayawardena, Abdul H Kanji, Ayush Khandelwal, Khoa Le, Vidhi Mody, Vrushti Mody, **Jarad Niemi**, Ariane Stark, Apurv Shah, Nutch

Wattanachit, Martha Zorn, and Nicholas Reich. (2022) The United States COVID-19 Forecast Hub dataset. *Scientific Data*, **9**(462). [url](#)

Estee Y Cramer, Evan L Ray, Velma K Lopez, Johannes Bracher, Andrea Brennen, Alvaro J Castro Rivadeneira, Aaron Gerding, Tilmann Gneiting, Katie H House, Yuxin Huang, Dasuni Jayawardena, Abdul H Kanji, Ayush Khandelwal, Khoa Le, Anja Muhlemann, **Jarad Niemi**, [256 other authors], Nicholas G Reich. (2022) Evaluation of individual and ensemble probabilistic forecasts of COVID-19 mortality in the US. *Proceedings of the National Academy of Sciences*, **119**(15), e2113561119. [url](#)

Matthew Stephenson, Lisa Schulte, Robert Klaver, and **Jarad Niemi**. (2022) Miniature temperature data loggers increase precision and reduce bias when estimating the daily survival rate for bird nests. *Journal of Field Ornithology*, **92**(4), 492–505. [url](#)

Mark Tomer, Sarah Porter, David James, Jessica Van Horn, and **Jarad Niemi**. (2021) Comparing riparian buffer design classification data among watersheds representing Iowa landscapes *Agrosystems, Geosciences, and Environment*, **4**(2), e20159. [url](#)

Adam Martin-Schwarze*, **Jarad Niemi**, and Philip Dixon. (2021) Joint modeling of distances and times in point-count surveys. *Journal of Agricultural, Biological, and Environmental Statistics*, **26**, 289–305. [url](#)

Ashley N. Dean, **Jarad B. Niemi**, John C. Tyndall, Erin W. Hodgson, and Matthew E. O’Neal. (2021) Developing a decision-making framework for insect pest management: a case study using soybean aphid (Hemiptera: Aphididae). *Pest Management Science*, **77**(2), 886–894. [url](#)

Nate Garton*, **Jarad Niemi**, and Alicia Carriquiry. (2020) Knot selection in sparse Gaussian processes with a variational objective. *Statistical Analysis and Data Mining*, **13**(4), 324–336. [url](#)

M.D. Tomer, J.D. Van Horn, S.A. Porter, D.E. James, and **J. Niemi**. (2020) Comparing agricultural conservation planning framework (ACPF) practice placements for runoff mitigation and controlled drainage among 32 watersheds representing Iowa landscapes. *Journal of Soil and Water Conservation*, **75**(4), 460–471. [url](#)

Colin Lewis-Beck*, **Jarad Niemi**, Petrutza Caragea, Brian Hornbuckle, Victoria Walker. (2020) Extracting agronomic information from SMOS vegetation optical depth in the US corn belt using a nonlinear hierarchical model. *Remote Sensing*, **12**(5), 827. [url](#)

E. A. Hines, M. R. Romoser, Z. E. Kiefer, A. F. Keating, L. H. Baumgard, **J. Niemi**, N. K. Gabler, J. F. Patience, G. Haberl, N. H. Williams, B. J. Kerr, K. J. Touchette, and J. W. Ross. (2019) The impact of dietary supplementation of arginine during gestation in a commercial swine herd: II. Offspring performance. *Journal of Animal Science*, **97**(9), 3626–3635. [url](#)

E. A. Hines, M. R. Romoser, Z. E. Kiefer, A. F. Keating, L. H. Baumgard, **J. Niemi**, N. K. Gabler, J. F. Patience, G. Haberl, N. H. Williams, B. J. Kerr, K. J. Touchette, and J. W. Ross. (2019) The impact of dietary supplementation of arginine during gestation in a commercial herd: I. Gilt reproductive performance. *Journal of Animal Science*, **97**(9), 3617–3625. [url](#)

Craig J. McGowan, Matthew Biggerstaff, Michael Johansson, Karyn M. Apfeldorf, Michal Ben-Nun, Logan Brooks, Matteo Convertino, Madhav Erraguntla, David C. Farrow, John Freeze, Saurav Ghosh, Sangwon Hyun, Sasikiran Kandula, Joceline Lega, Yang Liu, Nicholas Michaud*, Haruka Morita, **Jarad Niemi**, Naren Ramakrishnan, Evan L. Ray, Nicholas G. Reich, Pete Riley, Jeffrey Shaman, Ryan Tibshirani, Alessandro Vespignani, Qian Zhang, Carrie Reed & The Influenza Forecasting Working Group. (2019) Collaborative

efforts to forecast seasonal influenza in the United States, 2015–2016. *Scientific Reports*, **9**. [url](#)

Ignacio Alvarez* and **Jarad Niemi**. (2019) Fully Bayesian analysis of allele-specific RNA-seq data. *Mathematical Biosciences and Engineering*, **16**(6), 7751–7770. [url](#)

Nathaniel Garton* and **Jarad Niemi**. (2019) Multivariate temporal modeling of crime with dynamic linear models. *PLOS ONE*, **14**(7), e0218375. [url](#)

Will Landau*, **Jarad Niemi**, and Dan Nettleton. (2019) Fully Bayesian analysis of RNA-seq counts for the detection of gene expression heterosis. *Journal of the American Statistical Association*, **114**(526), 610–621. [url](#)

Adam Martin-Schwarze*, **Jarad Niemi**, and Philip Dixon. (2017) Assessing the impacts of time to detection distribution assumptions on detection probability estimation. *Journal of Agricultural, Biological, and Environmental Statistics*, **22**(4), 465–480. [url](#)

Lisa A. Schulte, **Jarad B. Niemi**, Matthew J. Helmers, Matt Liebman, J. G. Arbuckle, David E. James, Randall K. Kolka, Matthew E. O’Neal, Mark D. Tomer, John C. Tyndall, Heidi Asbjornsen, Pauline Drobney, Jeri Neal, Gary Van Ryswyk, and Chris Witte. (2017) Prairie strips improve biodiversity and the delivery of multiple ecosystem services from corn-soybean croplands *Proceedings of the National Academy of Sciences*, **114**(42), 11247–11252. [url](#)

Matthew Simpson*, **Jarad Niemi**, and Vivekananda Roy. (2017) Interweaving Markov chain Monte Carlo strategies for efficient estimation of dynamic linear models. *Journal of Computational and Graphical Statistics*, **26**:1, 152–159. [url](#)

Lisa Schulte, Anna MacDonald, **Jarad Niemi**, and Matthew Helmers. (2016) Prairie strips as a mechanism to promote land sharing by birds in industrial agricultural landscapes. *Agriculture, Ecosystems & Environment*, **220**, 55–63. [url](#)

Jarad Niemi, Eric Mittman*, Will Landau*, and Dan Nettleton. (2015) Empirical Bayes analysis of RNA-seq data for detection of gene expression heterosis. *Journal of Agricultural, Biological, and Environmental Statistics*, **20**(4), 614–628. [url](#)

David M. Brenner, Jack Dekker, **Jarad Niemi**, and Lisa Pfiffner. (2015) Medical oxygen concentrators for releasing seed dormancy. *Crop Science*, **55**(5), 2291–2293. [url](#)

Robert B. Gramacy, **Jarad Niemi**, and Robin Weiss. (2014) Massively parallel approximate Gaussian process regression. *SIAM/ASA Journal on Uncertainty Quantification*, **2**, 568–584. [url](#)

Daniel Sheinson*, **Jarad Niemi**, and Wendy Meiring. (2014) Comparison of the performance of particle filter algorithms applied to tracking of a disease epidemic. *Journal of Mathematical Biosciences*, **255**, 21–32. [url](#)

Jonathan Dess, Jennifer L. Momsen, **Jarad Niemi**, and Lisa Montplaisir. (2014) Student interpretations of phylogenetic trees in an introductory biology course. *CBE - Life Sciences Education*, **13**(4), 666–676. [url](#)

Jo Eidsvik, Benjamin A. Shaby, Brian J. Reich, Matthew Wheeler*, and **Jarad Niemi**. (2014) Estimation and prediction in spatial models with block composite likelihoods. *Journal of Computational and Graphical Statistics*, **23**(2), 295–315. [url](#)

Bernie J Daigle Jr, Min K Roh, Linda R Petzold, and **Jarad Niemi**. (2012) Accelerated maximum likelihood parameter estimation for stochastic biochemical systems. *BMC Bioinformatics*, **13**(68). [url](#)

David Banks, Gauri Datta, Alan Karr, James Lynch, **Jarad Niemi**, and Francisco Vera. (2012) Bayesian CAR models for syndromic surveillance on multiple data streams: theory and practice. *Information Fusion*, **13**(2), 105–116. [url](#)

Mike Ludkovski and **Jarad B. Niemi**. (2010) Optimal dynamic policies for influenza management. *Statistical Communications in Infectious Diseases*, **2**(1):5. [url](#)

Jarad B. Niemi and Mike West. (2010) Adaptive mixture modelling Metropolis methods for Bayesian analysis of non-linear state-space models. *Journal of Computational and Graphical Statistics*, **19**(2), 260–280. [url](#)

Quanli Wang, **Jarad Niemi**, Cheemeng Tan, Lingchong You, and Mike West. (2010) Image segmentation and dynamic lineage analysis in single-cell fluorescent microscopy. *Cytometry: Part A*, **77A**(1), 101–110. [url](#)

Jarad Niemi, Brad Carlin, and Jon Alexander. (2008) Contrarian strategies for NCAA tournament pools: a cure for March madness? *Chance*, **21**(1), 39–46. [url](#)

Michael J. Simmons, **Jarad B. Niemi**, Don-Felix Ryzek, Cecile Lamour, Joseph W. Goodman, Wojtek Kraszkiewicz, and Ryan Wolff. (2007) Cytotype regulation by telomeric *P* elements in *Drosophila melanogaster*: Interactions with *P* elements from M' strains. *Genetics*, **176**(4), 1957–1966.

Cheemeng Tan, Hao Song, **Jarad Niemi**, and Lingchong You. (2007) A synthetic biology challenge: making cells compute. *Molecular BioSystems*, **3**, 343–353.

Kevin J. Haley, Jeremy R. Stuart, John D. Raymond, **Jarad B. Niemi**, and Michael J. Simmons. (2005) Impairment of cytotypic regulation of *P*-element activity in *Drosophila melanogaster* by mutations in the Su(var)205 gene. *Genetics*, **171**, 583–595.

Jarad B. Niemi, John D. Raymond, Ryan Patrek, and Michael J. Simmons. (2004) Establishment and maintenance of the *P* cytotypic associated with telomeric *P* elements in *Drosophila melanogaster*. *Genetics*, **166**, 255–264.

Michael J. Simmons, John D. Raymond, **Jarad B. Niemi**, Jeremy R. Stuart, and Peter J. Merriman. (2004) The *P* cytotypic in *Drosophila melanogaster*: A maternally transmitted regulatory state of the germ line associated with telomeric *P* elements. *Genetics*, **166**, 243–254.

Michael J. Simmons, Kevin J. Haley, Craig D. Grimes, John D. Raymond, and **Jarad B. Niemi**. (2002) A hobo transgene that encodes the *P* element transposase in *Drosophila melanogaster*: Autoregulation and cytotypic control of transposase activity. *Genetics*, **161**, 195–204.

Book Chapters

Jarad B. Niemi and Gerald J. Niemi. (2015) Linear regression, model averaging, and Bayesian techniques for predicting chemical activities from structure. Ebook chapter on Advances in Mathematical Chemistry and Applications, Editors: Subhash C. Basak, Guillermo Restrepo and Jose Luis Villaveces, Bentham Publishers ([url](#))

Refereed Conference Proceedings

Ignacio Alvarez*, **Jarad Niemi**, and Matt Simpson*. (2014) Bayesian inference for a covariance matrix. *Proceedings for the Conference on Applied Statistics in Agriculture* ([url](#))

Michael Ludkovski and **Jarad Niemi** (2011) Optimal disease outbreak decisions using stochastic simulation. *Proceedings of the 2011 Winter Simulation Conference*, eds. S. Jain, R. R. Creasey, J. Himmelspack, K. P. White, and M. Fu.

Jarad Niemi, Meredith Smith, and David Banks. (2008) Test power for drug abuse surveillance. in *Biosurveillance and Biosecurity, Proceedings of BioSecure 2008, Lecture Notes in Computer Science*, eds. Daniel Zeng, Hsinchun Chen, Henry Rolka, and William B. Lober. pp. 131–142

Refereed Abstracts

Nicholas L. Michaud* and **Jarad Niemi**. (2016) A Bayesian Hierarchical Model for Estimating Influenza Epidemic Severity. ISDS Annual Conference Proceedings 2015. *Online Journal of Public Health Informatics* 8:1. ([url](#))

Nicholas L. Michaud*, Aaron Kite-Powell, and **Jarad Niemi**. (2016) CDCPlot: an Application for Viewing Weekly CDC MMWR Disease Count Data. *Online Journal of Public Health Informatics* 8:1. ([url](#))

Jarad Niemi. (2014) A Tool for Interactive Disease Outbreak Visualization, Detection, and Forecasting. *Online Journal of Public Health Informatics* 6:1. ([url](#))

Jarad Niemi and Michael Ludkovski. (2013) Tau-leaped particle learning. *Online Journal of Public Health Informatics* 5:1. ([url](#))

Jarad Niemi and Michael Ludkovski. (2011) Optimal sequential management decisions for measles outbreaks. *Emerging Health Threats Journal* 4. ([url](#))

Jarad Niemi. (2011) An exploratory analysis of the 2010 measles outbreak in Zimbabwe. *Emerging Health Threats Journal* 4. ([url](#))

Jarad B. Niemi, Michael D. Porter, and Brian J. Reich. (2008) Mixture likelihood ratio scan statistic for disease outbreak detection. *Advances in Disease Surveillance* 5:49. ([url](#))

Book reviews

Jarad B. Niemi. Statistics Done Wrong: The Woefully Complete Guide. *The American Statistician* 70(1), pg 125 [url](#)

Jarad B. Niemi. (2010) Dynamic Linear Models with R. *The American Statistician* 64(3), pg 268 [url](#)

Patent Applications

Tim W. Dake, **Jarad B. Niemi**, Don L. Hughes, Jeff J. Kester, Don B. Compton, Jon J. Calderas, Rich G. Schafermeyer, and Kevin P. Christmas. (2002) Compositions with high intensity sweetener having enhanced aqueous solubility and methods of their preparation. United States Patent Application 20030026872 <https://patentscope.wipo.int/search/en/detail.jsf?docId=W02002091856>

Other manuscripts

Luis Damiano* and **Jarad Niemi**. The RITAS Algorithm: a constructive yield monitor data processing algorithm. <https://arxiv.org/abs/2209.11313>

Will Landau* and **Jarad Niemi**. A fully Bayesian strategy for high-dimensional hierarchical modeling using massively parallel computing. <https://arxiv.org/abs/1606.06659>

Jarad Niemi and Andrew Gelman. (2011) Statistical graphics: making information clear – and beautiful. *Significance* 8, 135–137

Jarad B. Niemi and Matthew Wheeler*. (2011) Efficient Bayesian inference in stochastic chemical kinetic models using graphical processing units. <http://arxiv.org/abs/1101.4242>

Jarad B. Niemi. (2010) Evaluating individual player contributions to team offense and defense: a model based approach. *JSM Proceedings, Section on Statistics in Sports*. Vancouver, BC, Canada: American Statistical Association. 4914–4923.

Talks

- “Bayesian success in disease severity forecasting,” Best of Statistical Science (BOSS) 2026, 24 Apr 2026 (invited)
- “Specifying prior distributions in reliability applications,” SIAM Conference on Uncertainty Quantification (UQ26), 23 Mar 2026 (invited)
- “Bayesian methodology for disease severity forecasting,” University of Nebraska, Medical Center, 10 Oct 2025 (invited)
- “Bayesian methodology for disease severity forecasting,” Medical College of Wisconsin, 2 May 2025 (invited)
- “Gaussian Process Emulators with Functional Inputs for Water Erosion Prediction Project,” NCCC170, 15 Jun 2023 (contributed)
- “Building statistical emulators of computer models,” St. Olaf College, 14 Nov 2022 (invited)
- “Toward emulators for agricultural computer models,” Virginia Tech, 3 Nov 2022 (invited)
- “Advancing Scientific Practice through Agricultural Statistics,” Iowa State University, 26 Sep 2022 (promotion talk)
- “Computer Model Emulation in Agriculture,” NCCC170, 16 Jun 2022 (contributed)
- “Gaussian processes with functional length-scales for agricultural models,” Statistical Society of Canada Annual Meeting, 11 Jun 2021 (invited)
- “COVID-19 Forecast Hub,” Iowa State University Undergraduate STAT Club, 22 Oct 2020 (contributed)
- “COVID-19 forecast hub,” University of Iowa, 15 Oct 2020 (invited)
- “Emulation of Agricultural Production Systems sIMulator (APSIM),” Los Alamos National Labs, 23 Sep 2020 (invited)
- “Multivariate temporal modeling of crime with dynamic linear models,” Joint Statistical Meetings, 6 Aug 2020 (invited)
- “OAT knot selection for approximate Gaussian processes,” SIAM Conference on Uncertainty Quantification UQ20, March 2020 (invited; canceled due to COVID-19)
- “Sequential knot selection in sparse Gaussian processes,” Simon Fraser University, 10 Jan 2020 (invited)
- “Subfield Yield Analysis for Precision Agriculture,” Joint Statistical Meetings, 30 Jul 2019 (contributed)
- “Bayesian analysis in R,” Rgronomists/Lunchinators, 1 Feb 2019 (contributed)
- “Emulation of WEPP and DEP,” SAMSI MUMS Prediction Uncertainty and Extrapolation working group, 28 Nov 2018 (contributed)
- “Fully Bayesian Analysis of Hierarchical Count Regression Models,” Joint Statistical Meetings, 31 Jul 2018 (contributed)

“Fully Bayesian analysis for hierarchical count regression models,” University of California, Irvine, 28 Nov 2017 (invited)

“Fully Bayesian analysis of RNAseq data for gene expression heterosis detection,” University of Massachusetts - Amherst, 24 Mar 2017 (invited)

“Parallelized Markov chain Monte Carlo algorithms utilizing GPUs with an application to RNAseq data analysis,” 9th International Conference of the ERCIM WG on Computational and Methodological Statistics, University of Seville, Spain, 9-, 11 Dec 2016 (invited)

“Using information underlying missing data for improve estimation of NFL field goal kicker accuracy,” Midwest Sports Analytics Meetings, 19 Nov 2016 (contributed)

“Hierarchical modeling and data fusion,” Centers for Disease Control and Prevention Seasonal Influenza Forecasting Workshop, 31 Aug 2016 (invited)

“Massively parallel approximate Gaussian process regression,” Spring Research Conference, Illinois Institute of Technology, 26 May 2016 (invited)

“Stochastic dynamic models for low count observations,” Institute for Disease Modeling Symposium, 19 Apr 2016 (invited)

“Particle learning for low counts in disease outbreaks,” Institute for Disease Modeling, 18 Apr 2016 (invited)

“Bayesian analysis for heterosis detection in RNAseq data,” Department of Statistics, Colorado State University, 11 Dec 2015 (invited)

“Bayesian analysis for heterosis detection in RNAseq data,” Department of Applied Mathematics, University of Colorado-Boulder, 7 Dec 2015 (invited)

“Empirical Bayes analysis for detection of gene heterosis in RNAseq data,” Conference on Applied Statistics in Agriculture, Manhattan, KS, 27 Apr 2015 (contributed)

“Empirical Bayes identification of gene heterosis using RNAseq counts,” Animal Breeding and Genetics, Iowa State University, 28 Jan 2015 (invited)

“Massively parallel approximate Gaussian process regression,” Joint Statistical Meetings, 7 Aug 2014 (invited)

“Particle learning for low counts in disease outbreaks,” Department of Statistics, Iowa State University, 16 Sep 2013 (invited)

“Particle learning for sequential estimation and prediction of disease outbreaks,” 2013 ICSA-ISBS Joint Statistical Conference, 10 Jun 2013 (invited)

“Statistical methods for identifying gene expression heterosis,” Conference on Applied Statistics in Agriculture, Manhattan, KS, 30 Apr 2013 (contributed)

“Particle learning for low counts in disease outbreaks,” Midwest Statistics Research Colloquium, 15 Mar 2013 (invited)

“A computational approach to the sequential control problem,” Duke Department of Statistical Science 25th Anniversary Celebration, 20 Oct 2012 (invited)

“Rejection sampling on a graphical processing unit,” 11th World Meeting of the International Society for Bayesian Analysis, 28 Jun 2012 (invited)

“Determining optimal sequential disease outbreak interventions,” University of Iowa, Computational Epidemiology Seminar, 20 Jan 2012 (invited)

“Optimal sequential management decisions for measles outbreaks,” 10th International Society for Disease Surveillance 10th Annual Conference, 7 Dec 2011 (contributed)

“A sequential Monte Carlo primer,” Iowa State University, Department of Statistics, Computational Statistics working group, 12 Oct 2011 (invited)

“Statistical computing on graphical processing units,” Iowa State University, Department of Statistics, Computational Statistics working group, 28 Sep 2011 (invited)

“Time management,” University of California, Santa Barbara IGERT Career Development Seminar Series, 12 Apr 2011 (contributed)

“Optimal dynamic policies for influenza management,” Department of Statistics, Iowa State University, 24 Feb 2011 (invited)

“Optimal sequential management decisions for influenza outbreaks,” Department of Statistics and Applied Probability, University of California, Santa Barbara, 12 Jan 2011 (invited)

“Optimal sequential management decisions for influenza outbreaks,” 9th International Society for Disease Surveillance 9th Annual Conference, 2 Dec 2010 (contributed)

“Evaluating individual player contributions in basketball,” Joint Statistical Meetings, 4 Aug 2010 (contributed)

“A brief introduction to R,” University of California, Santa Barbara Quantitative Methods in Social Sciences seminar, 8 Apr 2010 (contributed)

“Early outbreak detection using syndromic surveillance networks,” Department of Biostatistics, University of California, Los Angeles, 19 Nov 2009 (invited)

“A sequential Monte Carlo primer,” Department of Statistics and Applied Probability, University of California, Santa Barbara, 21 Oct 2009 (invited)

“Adaptive mixture modeling Metropolis methods for state inference in nonlinear time series,” Joint Statistical Meetings, 3 Aug 2009 (invited)

“Computational methods for general state-space models,” Department of Mathematics and Statistics, University of New Mexico, 19 Feb 2009 (invited)

“Computational methods for general state-space models,” Department of Statistics, University of Michigan, 17 Feb 2009 (invited)

“Computational methods for general state-space models,” Department of Information, Risk, and Operations Management, University of Texas, Austin, 17 Feb 2009 (invited)

“Computational methods for general state-space models,” Department of Biostatistics, Johns Hopkins University, 6 Feb 2009 (invited)

“Computational methods for general state-space models,” Department of Biostatistics, University of California, Los Angeles, 28 Jan 2009 (invited)

“Computational methods for general state-space models,” Department of Statistics, Iowa State University, 26 Jan 2009 (invited)

“Computational approaches for general state-space models,” Department of Statistics and Applied Probability, University of California, Santa Barbara, 12 Jan 2009 (invited)

“Mixture likelihood ratio scan statistic for disease outbreak detection,” 7th Annual Meeting of the International Society for Disease Surveillance, 3 Dec 2008 (contributed)

“Test power for drug abuse surveillance,” BioSecure, 2 Dec 2008 (invited)

“Assessing the effectiveness of a national drug intervention policy,” Graduate Student Seminar Series, 17 Nov 2008 (contributed)

“Discrete-time models for intracellular processes in systems biology,” Graduate Student Research Day, Duke University, 2 Apr 2008 (contributed)

“Bayesian analysis in systems biology: advances and impact in single-cell dynamical networks,” Graduate Student Seminar Series, 25 Feb 2008 (contributed)

“Bayesian modeling and inference in single cell dynamic networks,” 39th Symposium on the Interface: Computing Science and Statistics, 26 May 2007 (contributed)

“Stochastic modelling and estimation in dynamic cellular networks,” 39th Symposium on the Interface: Computing Science and Statistics, 24 May 2007 (invited)

“Identifying and evaluating contrarian strategies for NCAA tournament pools,” 2006 Joint Statistical Meetings, 8 August, 8 Aug 2006 (contributed)

Posters

Jarad Niemi and Will Landau*. Fully Bayesian analysis of hierarchical count regression models applied to RNAseq. 2018 ISBA World Meeting, 26 Jun 2018

Nicholas L. Michaud*, Aaron Kite-Powell, and **Jarad Niemi**. CDCPlot: an Application for Viewing Weekly CDC MMWR Disease Count Data. 2015 International Society for Disease Surveillance Conference, 10 Dec 2015

Nicholas L. Michaud* and **Jarad Niemi**. A Bayesian Hierarchical Model for Estimating Influenza Epidemic Severity. Epidemics5, 3 Dec 2015

Will Landau* and **Jarad Niemi**. A Hierarchical Modeling Strategy for Identifying Gene Expression Heterosis using Parallel Computing with Graphics Processing Units (GPUs). Conference on Applied Statistics in Agriculture, Kansas State University, April 2015

Eric Mittman*, **Jarad Niemi**, and Dan Nettleton. Hierarchical Bayesian modeling of RNA-seq data with Stan. Conference on Applied Statistics in Agriculture, Kansas State University, April 2015

Jarad Niemi and Michael Ludkovski. Poisson-binomial state-space model for disease outbreaks with low counts. International Society for Bayesian Analysis World Meeting, 16 Jul 2014

Ignacio Alvarez* and **Jarad Niemi**. Bayesian Inference for covariance matrix. Conference on Applied Statistics in Agriculture, Manhattan, KS, 29 Apr 2014

Jarad Niemi. A Tool for Interactive Disease Outbreak Visualization, Detection, and Forecasting. 2013 International Society for Disease Surveillance Conference, 7 Dec 2013

A.L. MacDonald, L.A. Schulte Moore, M.J. Helmers, and **J.B. Niemi**. Bird response to native habitat strips integrated into agricultural fields. International Conference on Agricultural Biodiversity and Sustainability, Hokkaido, 29 Aug 2012

A.L. MacDonald, L.A. Schulte, M.J. Helmers, and **J.B. Niemi**. Testing a new conservation practice for agricultural landscapes: Bird response to prairie strips in row-cropped landscapes. 97th Annual Meeting of the Ecological Society of America, 7 Aug 2012

Jarad Niemi. An exploratory analysis of the 2010 measles outbreak in Zimbabwe. 2011 International Society for Disease Surveillance Conference, 7 Dec 2011

Jarad Niemi and Matthew Wheeler*. Parameter inference in stochastic chemical kinetic models on GPUs. MCM-Ski3: 4th International IMS/ISBA Joint Meeting, 6 Jan 2011

Jarad Niemi and Mike West. Nonlinear dynamic models for single-cell time-lapse microscopy. Duke Center for Systems Biology Retreat, 18 May 2009

Jarad Niemi. Adaptive mixture filtering: an alternative to particle filtering?. SAMSI Sequential Monte Carlo Kickoff Workshop, 8 Sep 2008

Jarad Niemi and Mike West. Bayesian parameter estimation for systems biological models of dynamic cellular networks. SAMSI Biosystems Modeling Workshop, 5 Mar 2007

Jarad Niemi and Mike West. Bayesian parameter estimation for systems biological models of dynamic cellular networks. 1st Annual Duke Systems Biology Symposium, 14 Sep 2006

Jarad Niemi and Mike West. Bayesian parameter estimation for systems biological models of dynamic cellular networks. 8th Valencia International Meeting on Bayesian Statistics, 5 Jun 2006

Jarad Niemi and Will Landau*. fbseq: An R package for fully Bayesian analysis of RNAseq data.

News interviews (hyperlinked)

[CNBC](#), [CBS Moneywatch](#), [MSN MoneyCentral](#), Slate ([2009](#), [2011](#), [2012](#), [2013](#)), The Duke Chronicle

Courses taught

Regular courses

At Iowa State University:		
Spring 2026	DS 4010	Data Science Capstone
Fall 2025	STAT 6150	Advanced Bayesian Methods
Spring 2025	STAT 4610X	Sports Analytics
	DS 4010	Data Science Capstone
Fall 2024	STAT 5870-1	Statistical Methods For Research Workers
Spring 2024	STAT 486/586	Introduction to Statistical Computing
	STAT 544	Bayesian Statistics
Spring 2023	STAT 486/586	Introduction to Statistical Computing
Fall 2022	STAT 330	Probability and Statistics for Computer Science
Spring 2022	HCI 522	Scientific Methods in Human Computer Interaction
Fall 2021	STAT 587-1	Statistical Methods For Research Workers
	STAT 615	Advanced Bayesian Methods
Spring 2021	STAT 330-2	Probability and Statistics for Computer Science
	STAT 587-3	Statistical Methods For Research Workers
Fall 2020	STAT 587-2	Statistical Methods For Research Workers
Spring 2019	STAT 544	Bayesian Statistics
	STAT 587C	Statistical Methods for Research Workers
Fall 2018	STAT 226	Introduction to Business Statistics
Spring 2018	STAT 544	Bayesian Statistics
	STAT 401C	Statistical Methods for Research Workers
Fall 2017	STAT 615	Advanced Bayesian Methods
Spring 2017	STAT 401D	Statistical Methods for Research Workers
	STAT 544	Bayesian Statistics
Fall 2016	STAT 330	Probability and Statistics for Computer Science
Spring 2016	STAT 544	Bayesian Statistics
Fall 2015	STAT 615	Advanced Bayesian Methods
Spring 2015	STAT 544	Bayesian Statistics
Fall 2014	STAT 401A	Statistical Methods for Research Workers
	STAT 330	Probability and Statistics for Computer Science
Spring 2014	STAT 544	Bayesian Statistics
Fall 2013	STAT 401A	Statistical Methods for Research Workers
	STAT 615	Advanced Bayesian Methods
Spring 2013	STAT 544	Bayesian Statistics
Fall 2012	STAT 401A	Statistical Methods for Research Workers
	HON 321E	Paradox
Spring 2012	STAT 401A	Statistical Methods for Research Workers
Fall 2011	STAT 615	Advanced Bayesian Methods
At University of California, Santa Barbara:		
Winter 2011	PSTAT 220B	Advanced Statistical Methods (GLMs)
Spring 2011	PSTAT 120C	Probability and Statistics
	PSTAT 230	Seminars and Projects in Statistical Consulting
Winter 2010	PSTAT 262	Applied Bayesian Time Series
Fall 2010	PSTAT 120B	Probability and Statistics
Spring 2010	PSTAT 120B	Probability and Statistics (estimation and testing)
	PSTAT 230	Seminar and Projects in Statistical Consulting

Short courses

Jan 2022	Data Management Workshop	Iowa State University	1-day
Dec 2016	Introduction to R for Biosurveillance	International Society for Disease Surveillance	1-day
Dec 2015	Introduction to R for Biosurveillance	International Society for Disease Surveillance	1-day
Dec 2014	Introduction to R for Biosurveillance	International Society for Disease Surveillance	1-day
Apr 2010	Sequential Monte Carlo methods	American Statistical Association, Albuquerque Chapter	1/2-day
Dec 2009	Introduction to statistical analysis in R	Natural Resources Research Institute	2-day

Grants

Iowa State University Presidential Interdisciplinary Research Initiative (ISU-PIRI), HYDRA: Hydration and Defense Resource Advancement, Role: Investigator, 1 Jul 2025 to 30 Jun 2028

USDA NIFA (#2020-68012-31824), Consortium for Cultivating Human and Naturally reGenerative Enterprises (C-CHANGE), Role: Co-PI, 1 Sep 2020 to 31 Aug 2025 [\$10,000,000; \$62,614 to Niemi]

Foundation for Food and Agriculture Research, Prairie Strips for Healthy Soils and Thriving Farms, Role: Co-PI, 1 Oct 2018 to 30 Sep 2021 [\$746,204.44; \$69,747 to Niemi]

Iowa State University Presidential Interdisciplinary Research Initiative (ISU-PIRI), Initiative for Cultivating Human And Naturally reGenerative Enterprises (I-CHANGE), Role: Co-PI, 1 Jun 2018 to 31 May 2021 [\$741,480; \$6,175 to Niemi]

National Institutes of Health, Hierarchical Modeling and Parallelized Bayesian Inference for the Analysis of RNAseq Data, Role: Co-PI, 1 Sep 2013 to 31 May 2017 [\$1,088,156; \$272,039 to Niemi]

Honors and Awards

Iowa State University Emerging Leaders Academy (AY25-26)

Iowa State University College of Liberal Arts & Sciences Inclusive Excellence (Team award including Ulrike Genschel, Laura Ziegler, and Ann Hawkins) (2025)

[Iowa State University College of Liberal Arts & Sciences Outstanding Achievement in Extension or Professional Practice](#) (2022)

[Iowa State University Interdisciplinary Team Research Award \(C-CHANGE\)](#) (2022)

[American Society of Agronomy Presidential Award \(NCCC170\)](#) (2021)

[Soil and Water Conservation Society Conservation Innovation Award \(STRIPS Team\)](#) (2020)

[Iowa State University College of Liberal Arts & Sciences Institutional Service Award](#) (2018)

[Iowa State University College of Agriculture and Life Sciences Team Award \(STRIPS Team\)](#) (2018)

Iowa State University Health Research Initiative on Infectious Diseases (PI Michael Cho, I was one of many investigators) [\$150,000] (2012)

International Society for Bayesian Analysis World Meeting Early Career Researchers Travel Grant [\$500] (2012)

Iowa State University College of Liberal Arts and Sciences grant for GPU cluster and RA support [\$88,000] (2011)

University of California Regents Junior Faculty Fellowship [\$7,500] (2011)

MCMSki3 conference travel support [\$650] (2011)

NVIDIA Academic Partnership Program award (2 NVIDIA C2050 GPUs) (2010)

Section on Bayesian Statistical Science Student Paper Competition winner [\$1,000] (2009)

International Society for Disease Surveillance Technical Contest 2nd place (2008)

National Science Foundation National Research Service Award Fellowship (AY03-04)

Memberships

American Association for the Advancement of Science

American Statistical Association

International Society for Bayesian Analysis

Service

Profession

- American Statistical Association Iowa Chapter, Treasurer (2026–present)
- National Science Foundation, Panel (2025, 2024, 2018)
- *Journal of Agricultural, Biological, and Environmental Statistics*, Associate Editor (2016–2026)
- Conference organization
 - * Conference on Predictive Inference in Sports, Conference Chair (2025)
 - * Conference on Applied Statistics in Agriculture and Natural Resources, Program Committee Member (2024)
 - * International Society for Disease Surveillance, Analytical Methods track chair (2013, 2012), Session chair (2010)
 - * Joint Statistical Meetings, Section on Statistical Computing session chair (2009)
- Other statistical community service
 - * [First Robotics Competition \(FRC\) Team Neutrino \(#3928\)](#), Mentor (2024–present)
 - * [Website, NCCC170](#), Development Lead (2022–present)
 - * First Lego League (FLL), Coach (2022–2024)
 - * Iowa State Science & Technology Fair, Judge (AY22-23)
 - * [International Society for Bayesian Analysis](#), Web Editor (2013–2015)
 - * [Professional YouTube channel](#) (2011–present)
 - * Significance Magazine, News Editor (2010–2012)
 - * [Professional website](#) (2009–present)
 - * [Cross Validated](#)

University service

- LAS CASTLE, Iowa State University (2020–2025), chair (2023–2025)
- Provost Working Group on Faculty Salary Equity Study, Iowa State University (2022–2024)
- Smart Technologies, C-CHANGE, Lead (2020–2024)
- STRIPS, Iowa State University, Data lead (2018–2024)
- Faculty Compensation Committee, Iowa State University (2016–2022)
- Transportation Advisory Council, Iowa State University (2015–2018)
- Faculty Senate, Iowa State University, departmental representative (2012–2018)
- Center for Research in Financial Mathematics and Statistics, University of California, Santa Barbara, member (AY10-11)
- Faculty Legislature, University of California, Santa Barbara, member (AY10-11)
- Quantitative Methods in the Social Sciences, University of California, Santa Barbara, core faculty (2009–2011)

Departmental/Program service

- Iowa State University Data Science Program
 - * Administrative Assistant Search Committee, co-chair joint with Psychology (2026)
 - * Data Science Curriculum Committee, chair (2024–present)
 - * Data Science Program, Director (2024–present)
- Iowa State University Department of Statistics
 - * Communications Specialist Search Committee, member (2025)
 - * Data Science Club, advisor (2025–present)
 - * Faculty Search Committee, member (AY25-26, AY21-22, AY18-19), chair (AY23-24)
 - * Statistical Consulting, Director (2025–present)
 - * Computation Advisory Committee, member (2023–present, 2011–2022), chair (2017–2019)
 - * Graduate Student Services Specialist Search Committee, member (2023, 2022)
 - * Graduate Curriculum and Exams Review, co-chair (2022–2024)
 - * Graduate Recruiting and Admissions Committee, director (2022–2025), member (2020–2026)
 - * Graduate Curriculum Review Committee, chair (AY21-22)
 - * Graduate Exams Review Committee, member (2021–2023, 2016–2019)
 - * Diversity, Equity, and Inclusion Committee, member (AY20-21)
 - * Sports Analytics Club, advisor (2020–present)
 - * Website Committee, member (2020–present), chair (2020–2024)
 - * Department Chair Search Advisory committee (AY18-19, 2012–2014)
 - * Snedecor Sustainability Committee, co-chair (2015–2017)
 - * Director of Statistical Consulting search committee (AY14-15)
 - * Statistics/BCB search committee (AY14-15)
 - * Computational Statistics working group, chair (2011–2014)
 - * STAT-ers, advisor (2011–2017)
 - * Social Committee (2011–2015)

Student Committees and Advisees

Table 1: Graduate students advised

Student	School	Department	Degree	Completed	Chair	Co-chair
Anna Diaz Farias	ISU	STAT	MS	2026	Chair	
David Gao	ISU	STAT	MS	2025	Chair	
Fredrick Osei	ISU	STAT	MS	2024	Co-chair	Danica Ommen
Spencer Wadsworth	ISU	STAT	MS	2022	Chair	
Luis Damiano	ISU	STAT	MS	2020	Chair	
Xiyuan Sun	ISU	STAT	MS	2019	Chair	
Gulzina Kuttubekova	ISU	STAT	MS	2019	Chair	
Nate Garton	ISU	STAT	MS	2018	Chair	
Katie Will	ISU	STAT	MS	2017	Chair	
Eric Mittman	ISU	STAT	MS	2015	Chair	
Ignacio Alvarez-Castro	ISU	STAT	MS	2014	Chair	
Adam Martin-Schwarze	ISU	STAT	MS	2014	Chair	
Casey Oliver	ISU	STAT	MS	2012	Co-chair	Dan Nettleton
Alexander Phillips	ISU	STAT	PhD	In progress	Co-chair	Pulong Ma
Aditya Ranade	ISU	STAT	PhD	2025	Chair	
Spencer Wadsworth	ISU	STAT	PhD	2024	Chair	
Luis Damiano	ISU	STAT	PhD	2023	Chair	
Nate Garton	ISU	STAT	PhD	2020	Co-chair	Alicia Carriquiry
Nehemias Ulloa	ISU	STAT	PhD	2019	Chair	
Colin Lewis-Beck	ISU	STAT	PhD	2018	Co-chair	Petrutza Caragea
Ignacio Alvarez-Castro	ISU	STAT	PhD	2017	Chair	
Eric Mittman	ISU	STAT	PhD	2017	Chair	
Adam Martin-Schwarze	ISU	STAT	PhD	2017	Co-chair	Philip Dixon
Will Landau	ISU	STAT	PhD	2016	Chair	
Nicholas Michaud	ISU	STAT	PhD	2016	Chair	
Matthew Simpson	ISU	STAT&ECON	PhD	2015	Co-chair	Gray Calhoun
Danny Sheinson	UCSB	PSTAT	MS	2011	Co-chair	Wendy Meiring
Matthew Wheeler	UCSB	PSTAT	MS	2010	Chair	
Danny Sheinson	UCSB	PSTAT	PhD	2014	Co-chair	Wendy Meiring

Table 2: Graduate student committees

Chair/Co-chair	STAT	Degree	Completed	In progress
Yes	Yes	MS	15	0
Yes	Yes	PhD	13	1
No	No	MS	14	0
No	No	PhD	28	6
No	Yes	MS	17	1
No	Yes	PhD	25	1

Table 3: Undergraduate Research Projects Supervised

Student	Year	Topic
Aridania Gerardo	2021	Bayesian Hierarchical Model for Reward Probabil...
Roger Castillo-Ramos	2021	Shiny App for Visualization of Fertilizer Effec...
Benedict Neo	2022	Development of WEPPR: an R interface to the WEP...
Benedict Neo	2023	Python Implementation of RITAS Yield-processing...
Ben Monroe	2023	Wrestling Data Analysis
Feifan (Fiona) Cao	2023	Empirical Study of Card Shuffling
Srilikitheswari (Lucky) Korrapati	2024–2025	Exploratory Statistics for Functions
Bhargav Yellepeddi	2025–2026	Statistical Modeling for Power Rating of FRC Ro...

Table 4: Students mentored

Student	Program	Institution	Years
Natalia da Silva	Preparing Future Faculty	Iowa State University	AY15-16
Will Landau	Preparing Future Faculty	Iowa State University	AY13-14